SIMATS ENGINEERING

ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Embedded Systems **COURSE CODE:** ECA1422

COURSE OUTCOME COVERED

CO1: ***Analyze and design complex and networked embedded systems using formal design methodologies, embedded architectures, CAN (BL5)***

Total Marks: 20

Sl. No	QUESTION BANK	BLOOMS LEVEL
14	Develop a detailed concept map representing the complete embedded product development cycle connecting concept definition,requirement analysis,system design,hardware-software co-design,prototyping,testing,validation,and product release and demonstrate how each phase contributes to product quality,time-to-market,and regulatory compliance in a real embedded systems.	BTL 5

Code of Conduct:

S. POOVITHA (192310183) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Evaluation Rubric (20 Marks per Question – Concept Map)

Criteria	Weightage	Level 4 (Exemplary) (5 marks)	Level 3 (Proficient) (4 marks)	Level 2 (Developing) (3 marks)	Level 1 (Beginning) (1-2 marks)
Concept Identification	25 %	25			
Linkages	25 %	25			
Organization	20 %	20			
Integration of Literature	20 %	20			
Clarity	10%	10			

100/100

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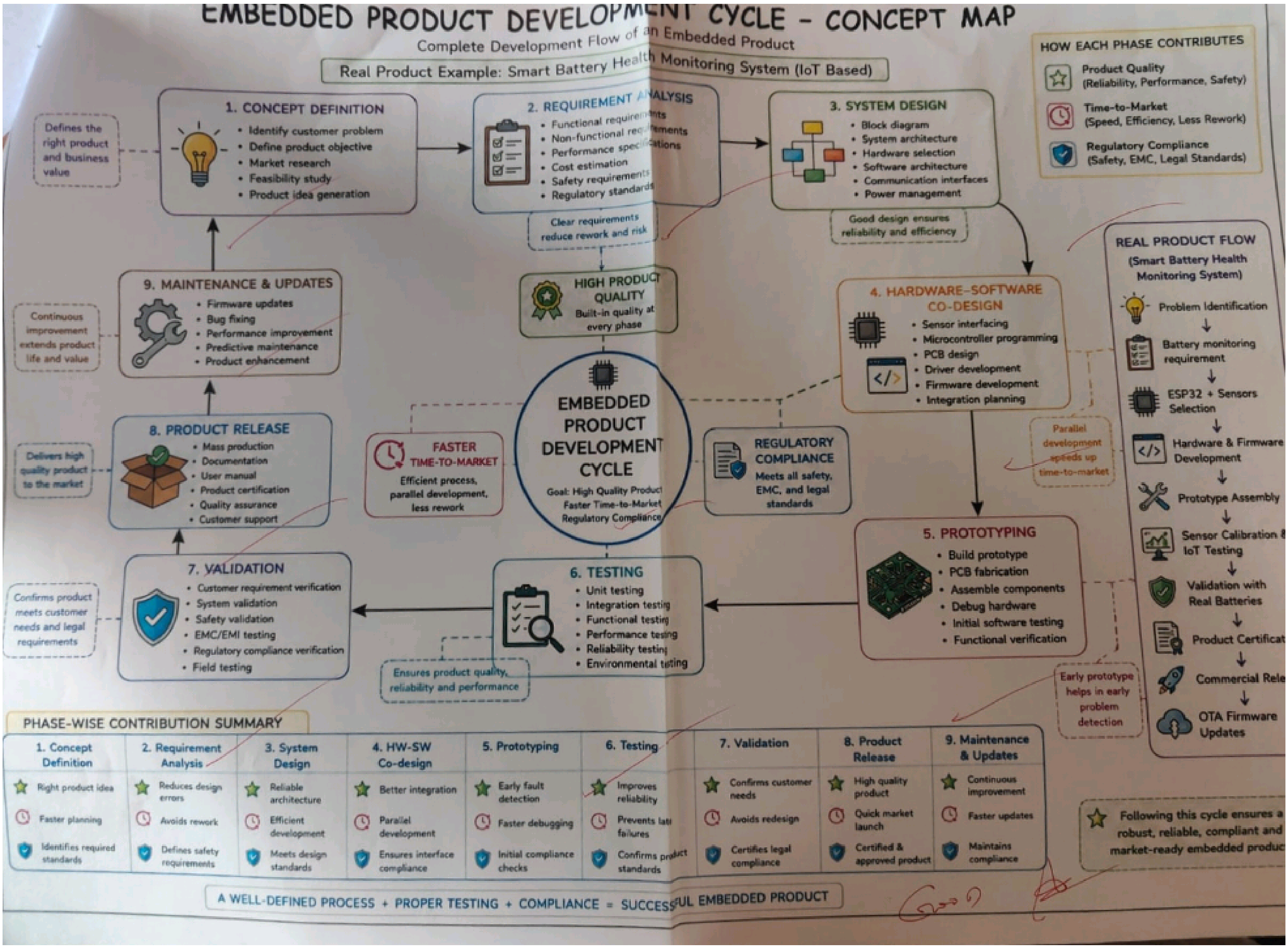
EMBEDDED PRODUCT DEVELOPMENT CYCLE - CONCEPT MAP
Complete Development Flow of an Embedded Product
Real Product Example: Smart Battery Health Monitoring System (IoT Based)
HOW EACH PHASE CONTRIBUTES
Product Quality (Reliability, Performance, Safety)
Time-to-Market (Speed, Efficiency, Less Rework)
Regulatory Compliance (Safety, EMC, Legal Standards)
Defines the right product and business value
1. CONCEPT DEFINITION
• Identify customer problem
• Define product objective
• Market research
• Feasibility study
• Product idea generation
2. REQUIREMENT ANALYSIS
• Functional requirements
• Non-functional requirements
• Performance specifications
• Cost estimation
• Safety requirements
• Regulatory standards
Clear requirements reduce rework and risk
3. SYSTEM DESIGN
• Block diagram
• System architecture
• Hardware selection
• Software architecture
• Communication interfaces
• Power management
Good design ensures reliability and efficiency
HIGH PRODUCT QUALITY
Built-in quality at every phase
Continuous improvement extends product life and value
9. MAINTENANCE & UPDATES
• Firmware updates
• Bug fixing
• Performance improvement
• Predictive maintenance
• Product enhancement
4. HARDWARE-SOFTWARE CO-DESIGN
• Sensor interfacing
• Microcontroller programming
• PCB design
• Driver development
• Firmware development
• Integration planning
Parallel development speeds up time-to-market
EMBEDDED PRODUCT DEVELOPMENT CYCLE
Goal: High Quality Product, Faster Time-to-Market, Regulatory Compliance
FASTER TIME-TO-MARKET
Efficient process, parallel development, less rework
REGULATORY COMPLIANCE
Meets all safety, EMC, and legal standards
Delivers high quality product to the market
8. PRODUCT RELEASE
• Mass production
• Documentation
• User manual
• Product certification
• Quality assurance
• Customer support
5. PROTOTYPING
• Build prototype
• PCB fabrication
• Assemble components
• Debug hardware
• Initial software testing
• Functional verification
Early prototype helps in early problem detection
Confirms product meets customer needs and legal requirements
7. VALIDATION
• Customer requirement verification
• System validation
• Safety validation
• EMC/EMI testing
• Regulatory compliance verification
• Field testing
6. TESTING
• Unit testing
• Integration testing
• Functional testing
• Performance testing
• Reliability testing
• Environmental testing
Ensures product quality, reliability and performance
REAL PRODUCT FLOW
(Smart Battery Health Monitoring System)
Problem Identification
Battery monitoring requirement
ESP32 + Sensors Selection
Hardware & Firmware Development
Prototype Assembly
Sensor Calibration & IoT Testing
Validation with Real Batteries
Product Certificat
Commercial Rele
OTA Firmware Updates
PHASE-WISE CONTRIBUTION SUMMARY
1. Concept Definition
Right product idea
Faster planning
Identifies required standards
2. Requirement Analysis
Reduces design errors
Avoids rework
Defines safety requirements
3. System Design
Reliable architecture
Efficient development
Meets design standards
4. HW-SW Co-design
Better integration
Parallel development
Ensures interface compliance
5. Prototyping
Early fault detection
Faster debugging
Initial compliance checks
6. Testing
Improves reliability
Prevents late failures
Confirms product standards
7. Validation
Confirms customer needs
Avoids redesign
Certifies legal compliance
8. Product Release
High quality product
Quick market launch
Certified & approved product
9. Maintenance & Updates
Continuous improvement
Faster updates
Maintains compliance
Following this cycle ensures a robust, reliable, compliant and market-ready embedded produc
A WELL-DEFINED PROCESS + PROPER TESTING + COMPLIANCE = SUCCESSFUL EMBEDDED PRODUCT
Good